Supplementary 1, selecting tool

Review question	"What is the impact of transitional care interventions between hospital and home on readmissions among older medical patients?"	
Reviewer Name:		Date:
Author name/study ID:		Year:
Title:		Journal:
	Include	**Exclude**
Patient population	- Elderly patients discharged from a general medical ward or emergency departments - Patients ≥ 65 years or mean age ≥ 75 years	- Patients ≤ 65 and mean age ≤ 75 - Not only medical patients - Discharging ward not specified - Only one patient group/diagnosis such as COPD, diabetes - Patients with psychiatric disorders or patients who are cognitively impaired - Other
Interventions	- Interventions in the transitional phase between hospital and home which examine the effect of the intervention on readmission rates between intervention group and control group	- No comparison/control group - Not an intervention study - Other
Comparators	- Usual or standard procedure	- Comparison between two interventions - Comparison with anything else than usual procedure - Other
Outcomes	- Readmission	- Readmission is not an outcome
Setting	- Intervention includes both pre- and post-discharge components	- Only pre-discharge components - Only post-discharge components - Other
Study design	- Quantitative studies	- Systematic reviews, meta-analysis, literature reviews - Case-reports or case-studies - Meeting abstract, study protocol - Other
Overall decision	- **Included**	- **Excluded**
Notes: Reason for exclusion:		

Supplementary 2, search string

Search string with free-text words used in PubMed

1	elder* (title)
2	aged (title)
3	old* (title)
4	geriatric* (title)
5	1 OR 2 OR 3 OR 4
6	"discharge intervention" (title)
7	transition* (title)
8	transition* care (title)
9	6 OR 7 OR 8
10	readmission* (text word)
11	re-admission* (text word)
12	rehospitalization* (text word)
13	re-hospitalization* (text word)
14	rehospitalisation* (text word)
15	re-hospitalisation* (text word)
16	10 OR 11 OR 12 OR 13 OR 14 OR 15
17	5 AND 9 AND 16
18	Filters: published in the last 10 years; Danish; English; Norwegian;

Search string with subject headings used in PubMed

1	Patient Discharge/organization and administration (MESH)
2	Transitional Care/organization and administration (MESH)
3	Transitional care/trends (MESH)
4	1 OR 2 OR 3
5	Readmission/organization and administration (MESH)
6	Patient Readmission/trends (MESH)
7	5 OR 6
8	Aged (MESH)
9	Aged, 80 and over (MESH)
10	4 AND 7 AND 9
12	Filters: published in the last 10 years; Danish; English; Norwegian;

Supplementary 3, Quality assessment tool and dictionary

Quality Assessment Tool for Quantitative Studies Dictionary



The purpose of this dictionary is to describe items in the tool thereby assisting raters to score study quality. Due to under-reporting or lack of clarity in the primary study, raters will need to make judgements about the extent that bias may be present. When making judgements about each component, raters should form their opinion based upon information contained in the study rather than making inferences about what the authors intended. Mixed methods studies can be quality assessed using this tool with the quantitative component of the study.

A) SELECTION BIAS

(Q1) Participants are more likely to be representative of the target population if they are randomly selected from a comprehensive list of individuals in the target population (score very likely). They may not be representative if they are referred from a source (e.g. clinic) in a systematic manner (score somewhat likely) or self-referred (score not likely).

(Q2) Refers to the % of subjects in the control and intervention groups that agreed to participate in the study before they were assigned to intervention or control groups.

B) STUDY DESIGN

In this section, raters assess the likelihood of bias due to the allocation process in an experimental study. For observational studies, raters assess the extent that assessments of exposure and outcome are likely to be independent. Generally, the type of design is a good indicator of the extent of bias. In stronger designs, an equivalent control group is present and the allocation process is such that the investigators are unable to predict the sequence.

Randomized Controlled Trial (RCT)

An experimental design where investigators randomly allocate eligible people to an intervention or control group. A rater should describe a study as an RCT if the randomization sequence allows each study participant to have the same chance of receiving each intervention and the investigators could not predict which intervention was next. If the investigators do not describe the allocation process and only use the words 'random' or 'randomly', the study is described as a controlled clinical trial.

See below for more details.

Was the study described as randomized?

Score YES, if the authors used words such as random allocation, randomly assigned, and random assignment.

Score NO, if no mention of randomization is made.

Was the method of randomization described?

Score YES, if the authors describe any method used to generate a random allocation sequence.

Score NO, if the authors do not describe the allocation method or describe methods of allocation such as alternation, case record numbers, dates of birth, day of the week, and any allocation procedure that is entirely transparent before assignment, such as an open list of random numbers of assignments.
If NO is scored, then the study is a controlled clinical trial.

Was the method appropriate?

Score YES, if the randomization sequence allowed each study participant to have the same chance of receiving each intervention and the investigators could not predict which intervention was next. Examples of appropriate approaches include assignment of subjects by a central office unaware of subject characteristics, or sequentially numbered, sealed, opaque envelopes.

Score NO, if the randomization sequence is open to the individuals responsible for recruiting and allocating participants or providing the intervention, since those individuals can influence the allocation process, either knowingly or unknowingly.

If NO is scored, then the study is a controlled clinical trial.

Controlled Clinical Trial (CCT)
An experimental study design where the method of allocating study subjects to intervention or control groups is open to individuals responsible for recruiting subjects or providing the intervention. The method of allocation is transparent before assignment, e.g. an open list of random numbers or allocation by date of birth, etc.

Cohort analytic (two group pre and post)
An observational study design where groups are assembled according to whether or not exposure to the intervention has occurred. Exposure to the intervention is not under the control of the investigators. Study groups might be non-equivalent or not comparable on some feature that affects outcome.

Case control study
A retrospective study design where the investigators gather 'cases' of people who already have the outcome of interest and 'controls' who do not. Both groups are then questioned or their records examined about whether they received the intervention exposure of interest.

Cohort (one group pre + post (before and after)
The same group is pretested, given an intervention, and tested immediately after the intervention. The intervention group, by means of the pretest, act as their own control group.

Interrupted time series
A study that uses observations at multiple time points before and after an intervention (the 'interruption'). The design attempts to detect whether the intervention has had an effect significantly greater than any underlying trend over time. Exclusion: Studies that do not have a clearly defined point in time when the intervention occurred and at least three data points before and three after the intervention

Other:
One time surveys or interviews

C) CONFOUNDERS

By definition, a confounder is a variable that is associated with the intervention or exposure and causally related to the outcome of interest. Even in a robust study design, groups may not be balanced with respect to important variables prior to the intervention. The authors should indicate if confounders were controlled in the design (by stratification or matching) or in the analysis. If the allocation to intervention and control groups is randomized, the authors must report that the groups were balanced at baseline with respect to confounders (either in the text or a table).

D) BLINDING

(Q1) Assessors should be described as blinded to which participants were in the control and intervention groups. The purpose of blinding the outcome assessors (who might also be the care providers) is to protect against detection bias.

(Q2) Study participants should not be aware of (i.e. blinded to) the research question. The purpose of blinding the participants is to protect against reporting bias.

E) DATA COLLECTION METHODS

Tools for primary outcome measures must be described as reliable and valid. If 'face' validity or 'content' validity has been demonstrated, this is acceptable. Some sources from which data may be collected are described below:

Self reported data includes data that is collected from participants in the study (e.g. completing a questionnaire, survey, answering questions during an interview, etc.).

Assessment/Screening includes objective data that is retrieved by the researchers. (e.g. observations by investigators).

Medical Records/Vital Statistics refers to the types of formal records used for the extraction of the data.

Reliability and validity can be reported in the study or in a separate study. For example, some standard assessment tools have known reliability and validity.

F) WITHDRAWALS AND DROP-OUTS

Score **YES** if the authors describe BOTH the numbers and reasons for withdrawals and drop-outs.

Score **NO** if either the numbers or reasons for withdrawals and drop-outs are not reported.

Score **NOT APPLICABLE** if the study was a one-time interview or survey where there was not follow-up data reported.

The percentage of participants completing the study refers to the % of subjects remaining in the study at the final data collection period in all groups (i.e. control and intervention groups).

G) INTERVENTION INTEGRITY

The number of participants receiving the intended intervention should be noted (consider both frequency and intensity). For example, the authors may have reported that at least 80 percent of the participants received the complete intervention. The authors should describe a method of measuring if the intervention was provided to all participants the same way. As well, the authors should indicate if subjects received an unintended intervention that may have influenced the outcomes. For example, co-intervention occurs when the study group receives an additional intervention (other than that intended). In this case, it is possible that the effect of the intervention may be over-estimated. Contamination refers to situations where the control group accidentally receives the study intervention. This could result in an under-estimation of the impact of the intervention.

H) ANALYSIS APPROPRIATE TO QUESTION

Was the quantitative analysis appropriate to the research question being asked?

An intention-to-treat analysis is one in which all the participants in a trial are analyzed according to the intervention to which they were allocated, whether they received it or not. Intention-to-treat analyses are favoured in assessments of effectiveness as they mirror the noncompliance and treatment changes that are likely to occur when the intervention is used in practice, and because of the risk of attrition bias when participants are excluded from the analysis.

Component Ratings of Study:

For each of the six components A – F, use the following descriptions as a roadmap.

A) SELECTION BIAS

Good: The selected individuals are very likely to be representative of the target population (Q1 is 1) **and** there is greater than 80% participation (Q2 is 1).

Fair: The selected individuals are at least somewhat likely to be representative of the target population (Q1 is 1 or 2); **and** there is 60 - 79% participation (Q2 is 2). 'Moderate' may also be assigned if Q1 is 1 or 2 and Q2 is 5 (can't tell).

Poor: The selected individuals are not likely to be representative of the target population (Q1 is 3); **or** there is less than 60% participation (Q2 is 3) **or** selection is not described (Q1 is 4); and the level of participation is not described (Q2 is 5).

B) DESIGN

Good: will be assigned to those articles that described RCTs and CCTs.

Fair: will be assigned to those that described a cohort analytic study, a case control study, a cohort design, or an interrupted time series.

Weak: will be assigned to those that used any other method or did not state the method used.

C) CONFOUNDERS

Good: will be assigned to those articles that controlled for at least 80% of relevant confounders (Q1 is 2); **or** (Q2 is 1).

Fair: will be given to those studies that controlled for 60 – 79% of relevant confounders (Q1 is 1) **and** (Q2 is 2).

Poor: will be assigned when less than 60% of relevant confounders were controlled (Q1 is 1) **and** (Q2 is 3) **or** control of confounders was not described (Q1 is 3) **and** (Q2 is 4).

D) BLINDING

Good: The outcome assessor is not aware of the intervention status of participants (Q1 is 2); **and t**he study participants are not aware of the research question (Q2 is 2).

Fair: The outcome assessor is not aware of the intervention status of participants (Q1 is 2); **or** the study participants are not aware of the research question (Q2 is 2).

Poor: The outcome assessor is aware of the intervention status of participants (Q1 is 1); **and** the study participants are aware of the research question (Q2 is 1); **or b**linding is not described (Q1 is 3 and Q2 is 3).

E) DATA COLLECTION METHODS

Good: The data collection tools have been shown to be valid (Q1 is 1); **and** the data collection tools have been shown to be reliable (Q2 is 1).

Fair: The data collection tools have been shown to be valid (Q1 is 1); **and** the data collection tools have not been shown to be reliable (Q2 is 2) **or** reliability is not described (Q2 is 3).

Poor: The data collection tools have not been shown to be valid (Q1 is 2) **or** both reliability and validity are not described (Q1 is 3 and Q2 is 3).

F) WITHDRAWALS AND DROP-OUTS - a rating of:

Good: will be assigned when the follow-up rate is 80% or greater (Q1 is 1 and Q2 is 1).

Fair: will be assigned when the follow-up rate is 60 – 79% (Q2 is 2) **OR** Q1 is 4 or Q2 is 5.

Poor: will be assigned when a follow-up rate is less than 60% (Q2 is 3) or if the withdrawals and drop-outs were not described (Q1 is No or Q2 is 4).

Not Applicable: if Q1 is 4 or Q2 is 5.

QUALITY ASSESSMENT TOOL FOR QUANTITATIVE STUDIES



COMPONENT RATINGS

A) SELECTION BIAS

(Q1) Are the individuals selected to participate in the study likely to be representative of the target population?

1 Very likely
2 Somewhat likely
3 Not likely
4 Can't tell

(Q2) What percentage of selected individuals agreed to participate?

1 80 - 100% agreement
2 60 – 79% agreement
3 less than 60% agreement
4 Not applicable
5 Can't tell

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

B) STUDY DESIGN

Indicate the study design

1 Randomized controlled trial
2 Controlled clinical trial
3 Cohort analytic (two group pre + post)
4 Case-control
5 Cohort (one group pre + post (before and after))
6 Interrupted time series
7 Other specify _____
8 Can't tell

Was the study described as randomized? If NO, go to Component C.

No Yes

If Yes, was the method of randomization described? (See dictionary)

No Yes

If Yes, was the method appropriate? (See dictionary)

No Yes

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

C) CONFOUNDERS

(Q1) Were there important differences between groups prior to the intervention?

1 Yes
2 No
3 Can't tell

The following are examples of confounders:

1 Race
2 Sex
3 Marital status/family
4 Age
5 SES (income or class)
6 Education
7 Health status
8 Pre-intervention score on outcome measure

(Q2) If yes, indicate the percentage of relevant confounders that were controlled (either in the design (e.g. stratification, matching) or analysis)?

1 80 – 100% (most)
2 60 – 79% (some)
3 Less than 60% (few or none)
4 Can't Tell

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

D) BLINDING

(Q1) Was (were) the outcome assessor(s) aware of the intervention or exposure status of participants?

1 Yes
2 No
3 Can't tell

(Q2) Were the study participants aware of the research question?

1 Yes
2 No
3 Can't tell

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

E) DATA COLLECTION METHODS

(Q1) Were data collection tools shown to be valid?

1 Yes
2 No
3 Can't tell

(Q2) Were data collection tools shown to be reliable?

1 Yes
2 No
3 Can't tell

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

F) WITHDRAWALS AND DROP-OUTS

(Q1) Were withdrawals and drop-outs reported in terms of numbers and/or reasons per group?

1 Yes
2 No
3 Can't tell
4 Not Applicable (i.e. one time surveys or interviews)

(Q2) Indicate the percentage of participants completing the study. (If the percentage differs by groups, record the lowest).

1 80 -100%
2 60 - 79%
3 less than 60%
4 Can't tell
5 Not Applicable (i.e. Retrospective case-control)

RATE THIS SECTION	STRONG	MODERATE	WEAK	
See dictionary	1	2	3	Not Applicable

G) INTERVENTION INTEGRITY

(Q1) What percentage of participants received the allocated intervention or exposure of interest?

1 80 -100%
2 60 - 79%
3 less than 60%
4 Can't tell

(Q2) Was the consistency of the intervention measured?

1 Yes
2 No
3 Can't tell

(Q3) Is it likely that subjects received an unintended intervention (contamination or co-intervention) that may influence the results?

4 Yes
5 No
6 Can't tell

H) ANALYSES

(Q1) Indicate the unit of allocation (circle one)

community organization/institution practice/office individual

(Q2) Indicate the unit of analysis (circle one)

community organization/institution practice/office individual

(Q3) Are the statistical methods appropriate for the study design?

1 Yes
2 No
3 Can't tell

(Q4) Is the analysis performed by intervention allocation status (i.e. intention to treat) rather than the actual intervention received?

1 Yes
2 No
3 Can't tell

GLOBAL RATING

COMPONENT RATINGS

Please transcribe the information from the gray boxes on pages 1-4 onto this page. See dictionary on how to rate this section.

A	**SELECTION BIAS**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
B	**STUDY DESIGN**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
C	**CONFOUNDERS**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
D	**BLINDING**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
E	**DATA COLLECTION METHOD**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
F	**WITHDRAWALS AND DROPOUTS**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	Not Applicable

GLOBAL RATING FOR THIS PAPER (circle one):

1	STRONG	(no WEAK ratings)
2	MODERATE	(one WEAK rating)
3	WEAK	(two or more WEAK ratings)

With both reviewers discussing the ratings:

Is there a discrepancy between the two reviewers with respect to the component (A-F) ratings?

No Yes

If yes, indicate the reason for the discrepancy

1	Oversight
2	Differences in interpretation of criteria
3	Differences in interpretation of study

Final decision of both reviewers (circle one):

1	**STRONG**
2	**MODERATE**
3	**WEAK**

Supplementary 4, exclusion of full text

Altfeld S.J., Shier G.E., Rooney M., Johnson T.J., Golden R.L., Karavolos K., Avery E., Nandi V. & Perry, A.J. 2013, "Effects of an enhanced discharge planning intervention for hospitalized older adults: a randomized trial.", The Gerontologist, vol. 53, no. 3, pp. 430-440. **Reason for exclusion:** not only medical patients.

Alvarez R, Ginsburg J, Grabowski J, Post S, Rosenberg W. The Social Work Role in Reducing 30-Day Readmissions: The Effectiveness of the Bridge Model of Transitional Care. J Gerontol Soc Work 2016 Apr;59(3):222-227. **Reason for exclusion:** not an intervention study.

Arbaje A.I., Maron D.D., Yu Q., Wendel V.I., Tanner E., Boult C., Eubank K.J. & Durso, S.C. 2010, "The geriatric floating interdisciplinary transition team.", Journal of the American Geriatrics Society, vol. 58, no. 2, pp. 364-370. **Reason for exclusion:** readmission is not an outcome.

Arnold ME, Buys L, Fullas F. Impact of pharmacist intervention in conjunction with outpatient physician follow-up visits after hospital discharge on readmission rate (Reprinted from Journal of Health-System Pharmacy Residents JHPR, vol 2, issue 4, 2014). American Journal of Health-System Pharmacy 2015 JUN 1;72(11):S36-S42. **Reason for exclusion:** only post-discharge elements, not only medical patients.

Arevalo-Flechas, L.C., Noel, P., Rottman-Sagebiel, R., Cupples, N., Conde, A., Sanchez-Reilly, S., Powers, B. & Espinoza, S. 2016, "Presence of a Caregiver Associated with Increased Odds of Readmission in a Cohort of Older Adults Included in a Transitional Care Intervention", Journal of the American Geriatrics Society, vol. 64, pp. S141-S141. **Reason for exclusion:** meeting abstract.

Bellon, J.E., Bilderback, A., Ahuja-Yende, N.S., Wilson, C., Altieri Dunn, S.C., Brodine, D. & Boninger, M.L. 2018, "University of Pittsburgh Medical Center Home Transitions Multidisciplinary Care Coordination Reduces Readmissions for Older Adults", Journal of the American Geriatrics Society. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Burt S, Berry D, Quackenbush P. Implementation of Transition in Care and Relationship Based Care to Reduce Preventable Rehospitalizations. Home Healthc Now 2015 Jul-Aug;33(7):390-393. **Reason for exclusion:** not an intervention study.

Cheen MHH, Goon CP, Ong WC, Lim PS, Wan CN, Leong MY, et al. Evaluation of a care transition program with pharmacist-provided home-based medication review for elderly Singaporeans at high risk of readmissions. Int J Qual Health Care 2017 Apr 1;29(2):200-205. **Reason for exclusion:** not only medical patients, only post-discharge elements, mean age < 75 year or age < 65 years.

Damiani G, Federico B, Venditti A, Sicuro L, Rinaldi S, Cirio F, et al. Hospital discharge planning and continuity of care for aged people in an Italian local health unit: does the care-home model reduce hospital readmission and mortality rates? BMC Health Serv Res 2009 Feb 4;9:22-6963-9-22. **Reason for exclusion:** not able to obtain data from the authors.

Dedhia P., Kravet S., Bulger J., Hinson T., Sridharan A., Kolodner K., Wright S. & Howell, E. 2009, "A quality improvement intervention to facilitate the transition of older adults from three hospitals back to their homes.", Journal of the American Geriatrics Society, vol. 57, no. 9, pp. 1540-1546. **Reason for exclusion:** only pre-discharge elements.

Dizon, M.L. & Reinking, C. 2017, "Reducing Readmissions: Nurse-Driven Interventions in the Transition of Care From the Hospital", Worldviews on evidence-based nursing, vol. 14, no. 6, pp. 432-439. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Edmans J, Bradshaw L, Franklin M, Gladman J, Conroy S. Specialist geriatric medical assessment for patients discharged from hospital acute assessment units: randomised controlled trial. BMJ 2013 Oct 8;347:f5874. **Reason for exclusion:** not able to obtain data from the authors.

Farrell T.W., TomoaiaCotisel A., Scammon D.L., Brunisholz K., Kim J., Day J., Gren L.H., Wallace S., Gunning K., Tabler J. & Magill, M.K. 2015, "Impact of an integrated transition management program

in primary care on hospital readmissions.", Journal for healthcare quality : official publication of the National Association for Healthcare Quality, vol. 37, no. 1, pp. 81-92. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Field, T.S., Ogarek, J., Garber, L., Reed, G. & Gurwitz, J.H. 2015, "Association of early post-discharge follow-up by a primary care physician and 30-day rehospitalization among older adults", Journal of general internal medicine, vol. 30, no. 5, pp. 565-571. **Reason for exclusion:** not an intervention study.

Gardner R, Li Q, Baier RR, Butterfield K, Coleman EA, Gravenstein S. Is implementation of the care transitions intervention associated with cost avoidance after hospital discharge? J Gen Intern Med 2014 Jun;29(6):878-884. **Reason for exclusion:** readmission is not an outcome.

Gurwitz, J.H., Field, T.S., Ogarek, J., Tjia, J., Cutrona, S.L., Harrold, L.R., Gagne, S.J., Preusse, P., Donovan, J.L., Kanaan, A.O., Reed, G. & Garber, L. 2014, "An Electronic Health Record-Based Intervention to Increase Follow-Up Office Visits and Decrease Rehospitalization in Older Adults", Journal of the American Geriatrics Society, vol. 62, no. 5, pp. 865-871. **Reason for exclusion:** not only medical patients.

Hamar, B., Rula, E.Y., Wells, A.R., Coberley, C., Pope, J.E. & Varga, D. 2016, "Impact of a Scalable Care Transitions Program for Readmission Avoidance", American Journal of Managed Care, vol. 22, no. 1, pp. 28-34. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Harrison, J.D., Auerbach, A.D., Quinn, K., Kynoch, E. & Mourad, M. 2014, "Assessing the impact of nurse post-discharge telephone calls on 30-day hospital readmission rates", JGIM: Journal of General Internal Medicine, vol. 29, no. 11, pp. 1519-1525. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Hendrix C., Tepfer S., Forest S., Ziegler K., Fox V., Stein J., McConnell E.S., Hastings S.N., Schmader K. & ColonEmeric, C. 2013, "Transitional Care Partners: a hospital-to-home support for older adults and their caregivers.", Journal of the American Association of Nurse Practitioners, vol. 25, no. 8, pp. 407-414. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Heppenstall C.P., Hanger H.C., Wilkinson T.J. & Dhanak, M. 2018, "Telephone discharge support for frail, vulnerable older people discharged from hospital: Impact on readmission rates - Participant and general practitioner feedback.", Australasian journal on ageing, vol. 37, no. 2, pp. 107-112. **Reason for exclusion:** only post-discharge elements.

Hitch B, Parlier AB, Reed L, Galvin SL, Fagan EB, Wilson CG. Evaluation of a Team-Based, Transition-of-Care Management Service on 30-Day Readmission Rates. N C Med J 2016 Mar-Apr;77(2):87-92. **Reason for exclusion:** not only medical patients, mean age < 75 year or age < 65 years.

Hung D. & Leidig, R.C. 2015, "Implementing a transitional care program to reduce hospital readmissions among older adults.", Journal of nursing care quality, vol. 30, no. 2, pp. 121-129. **Reason for exclusion:** not an intervention study.

Hung D., Truong Q., Yakir M. & Nicosia, F. 2018, "Hospital-Community Partnerships to Aid Transitions for Older Adults: Applying the Care Transitions Framework.", Journal of nursing care quality, vol. 33, no. 3, pp. 221-228. **Reason for exclusion:** not an intervention study.

Hwang, U., Dresden, S.M., Rosenberg, M.S., Garrido, M.M., Loo, G., Sze, J., Gravenor, S., Courtney, D.M., Kang, R., Zhu, C.W., Vargas-Torres, C., Grudzen, C.R., Richardson, L.D. & The GEDI, W.I. 2018, "Geriatric Emergency Department Innovations: Transitional Care Nurses and Hospital Use", Journal of the American Geriatrics Society, vol. 66, no. 3, pp. 459-466. **Reason for exclusion:** only pre-discharge elements.

Jeangsawang, N., Malathum, P., Panpakdee, O., Brooten, D. & Nityasuddhi, D. 2012, "Comparison of Outcomes of Discharge Planning and Post-Discharge Follow-up Care, provided by Advanced Practice,

Expert-by-experience, and Novice Nurses, to Hospitalized Elders with Chronic Healthcare Conditions", Pacific Rim International Journal of Nursing Research, vol. 16, no. 4, pp. 343-360. **Reason for exclusion:** not an intervention study.

Jenq G.Y., Doyle M.M., Belton B.M., Herrin J. & Horwitz, L.I. 2016, "Quasi-experimental evaluation of the effectiveness of a large-scale readmission reduction program.", JAMA Internal Medicine, vol. 176, no. 5, pp. 681-690. **Reason for exclusion:** not able to obtain data from the authors.

Kane R.L., Huckfeldt P., Tappen R., Engstrom G., Rojido C., Newman D., Yang Z. & Ouslander, J.G. 2017, "Effects of an intervention to reduce hospitalizations from nursing homes: A randomized implementation trial of the INTERACT program.", JAMA Internal Medicine, vol. 177, no. 9, pp. 1257-1264. **Reason for exclusion:** only post-discharge elements.

Kirkham, H., Clark, B., Paynter, J., Lewis, G. & Duncan, I. 2014, "The effect of a collaborative pharmacist-hospital care transition program on the likelihood of 30-day readmission", American Journal of Health-System Pharmacy, vol. 71, no. 9, pp. 739-745. **Reason for exclusion:** not only medical patients, mean age < 75 year or age < 65 years.

Latour, C.H., Bosmans, J.E., van Tulder, M.W., de Vos, R., Huyse, F.J., de Jonge, P., van Gemert, L.A. & Stalman, W.A. 2007, "Cost-effectiveness of a nurse-led case management intervention in general medical outpatients compared with usual care: an economic evaluation alongside a randomized controlled trial", Journal of psychosomatic research, vol. 62, no. 3, pp. 363-370. **Reason for exclusion:** mean age < 75 year or age < 65 years.

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Supplementary 5, intervention characteristics

Study, year	Title	Description of intervention components	Providers	Context	Length of support	Treatment of comparison group
Buurman et al.[24] **2016**	Transitional Care Bridge Program Intervention	Pre-discharge: *≤ 48 hours after hospital admission*: - participants received a systematic CGA. - Care and treatment plan with prioritazion by patient and geriatric team -Multidisciplinary care provided by geriatric team Bridging: - In-person handover of CGA and care treatment plan by geriatric team to CCRN - Visit of CCRN to participant Post-discharge: *≤ 2 days after hospital discharge:* - Home visit or visit to the NH, with medication reconciliation; initiate home care; check for activities of daily living and availability of adequate help; target geriatric conditions; contact GP or NH physician *2 weeks after hospital discharge*: - Home visit or visit to the NH, evaluation of hospital-based care and treatment plan, goal setting by participant, protocol adherence for incident and prevalent geriatric conditions *6 weeks after hospital discharge*: - Home visit or visit to the NH, focusing on the proxy and caregiver burden; further actions on geriatric conditions *12 weeks after hospital discharge*: - Home visit or visit to the NH, focusing on care needs for incident and prevalent geriatric conditions *24 weeks after hospital discharge*: - Home visit or visit to the NH, focusing on care needs; evaluation of geriatric conditions; handover to GP	Geriatric team A community care registered nurse (CCRN)	Three hospitals with affiliated home care organizations in the Netherlands	From ≤ 48 hours after hospital admission to 24 weeks after hospital discharge	CGA-Only Control Arm (usual care): - a systematic CGA - care and treatment plan with prioritazion by patient and geriatric team -multidisciplinary care provided by geriatric team
Chow et al.[22]	The nurse case management intervention	Two intervention arms: 1. "Home Visit" 2. "Call"	Senior year nursing students	Patients were recruited from the medical	From enrolment until 4 weeks after discharge	Social calls twice within 4 weeks.

2014		Pre-discharge in both "Home Visit" and "Call" arms: - Pre-discharge assessment: the Problem Classification Scheme, Intervention Scheme and Problem Rating Scale for Outcomes (e.g. analysis of the barriers and developed mutual goals with the patients to ensure that they were able to perform health maintenance activities, such as nutrition, monitoring of symptoms and medication adherence) Bridging: None Post-discharge - Home Visit arm: - First visit within 72 hours of discharge. - Week 2: a telephone call to evaluate the interventions and advice given during the home visit - Week 3: home visit to examine whether patients were able to continue managing their health needs after hospital discharge. - Week 4: final telephone call in the fourth week post discharge to remind patients about adherence behaviors and to motivate and support them before concluding the interventions Post-discharge – Call arm: - Week 1: first telephone call based on the patients' needs, which had been identified prior to discharge. - Week 2. call the patients: referred to the mutual goals developed by the NCM and the patients to continue with the interventions. - Week 3. call the patients referred to the mutual goals developed by the NCM and the patients to continue with the interventions. - Week 4: closing call to motivate and support the patients in maintaining self-management behaviors	Nurse Case Managers (NCM)	department of a 1700-bedded acute, general regional hospital in Hong Kong		No assessments and related interventions were conducted to the control group.
Courtney et al.[25] **2009**	Older Hospitalised Patients' Discharge Planning and In-home	Pre-discharge: *Exercise components*: - Muscle stretching, balance training, walking for endurance, and muscle strengthening using resistance exercises	A registered nurse (RN) A physio-therapist	A tertiary referral hospital in Brisbane, Australia	From enrolment until 6 months after discharge	Control group received the routine care, discharge planning, and rehabilitation advice normally provided. If

	Follow-up Protocol (OHP-DP)	*Nurse components:* - The nurse visited daily during participants' hospital stays to address concerns, facilitate the exercise program, and oversee discharge planning - Developed a transitional care plan covering the areas of functional ability and need for assistance with activities of daily living, post-discharge treatments and follow-up care, social support, chronic disease management plans and information, medication information, community services, and assistance with the exercise program. <u>Bridging</u>: - Written guidelines were provided on post-discharge management, including diagrams and specific instructions for their exercise program <u>Post-discharge</u>: - Within 48 hours of discharge, the nurse undertook a home visit to assess availability of support, address transitional concerns, provide advice and support, and ensure that the exercise program could be safely undertaken at home. - Extra home visits were provided if required - Weekly follow-up telephone calls were provided for 4 weeks, followed by monthly follow-up for a further 5 months				in-home follow-up was necessary, it was organized in the routine manner (e.g., referral to community health services)
Finlayson et al.[23] **2018**		Three intervention arms: 1. "Exercise" 2. "Nurse Home visit and Telephone follow-up" (N-HaT) 3. "Exercise and Nurse Home Visit and Telephone follow up" (ExN-HaT) <u>Exercise arm:</u> *Pre-discharge*: - Within 72 hours of admission, a physiotherapist will visit measure ADL and conduct performance tests of balance and gait. This information will be used to plan individualised exercise programs designed to improve strength, stability, coordination, endurance, mobility, and improve self confidence with respect to ADL. The exercise prescription will be developed by the patient, care givers, doctors, and ward nurses. -Physiotherapist will provide a pedometer	Advanced Practice Gerontic Nurse (APGN) A physiotherapist or exercise physiologist	Two tertiary metropolitan hospitals in Australia	From enrolment until 24 weeks after discharge	Routine hospital and follow-up care as provided by the health service. This involved a needs assessment by the hospital health staff, discharge planning, and referrals for follow up services as appropriate.

		together with simple verbal and written instructions regards usage and journaling of activity levels - APGN will visit patients 2nd daily thereafter until discharge to establish and implement the program, monitor progress, and modify transitional care plan if required *Bridging:* None *Post-discharge:* - An exercise physiologist conducts 6 weekly in home visits to reassess the patient's physical measures and functional capacity, evaluate progress with the exercise program, and reset program goals accordingly <u>N-HaT arm:</u> *Pre-discharge*: - Within 72 hours of admission, an APGN will undertake a health assessment, and prepare a comprehensive transitional care plan. This process includes discussing the planned discharge plan with patient's caregivers, doctor, and ward nurses to individualise the plan - APGN will visit patients 2nd daily thereafter until discharge to establish and implement the program, monitor progress, and modify transitional care plan if required. *Bridging:* None *Post-discharge:* - Within 48 hours after discharge, an APGN conducts a home visit. This home visit is undertaken to 1) identify sufficient caregiver support is available at home, 2) assess the home environment is safe, 3) determine the patient has the required medications and dressings (if required), 4) ensure the patient (and caregiver) fully understand their medication and treatment regimes; 5) reinforce and further explain exercise program and use of pedometer and journal in the home, and 6) provide advice and support to the caregiver				

		-Additional home visits are available by APGN if required - over the initial 4 week period after discharge, a APGN provides weekly telephone follow-up calls -APGN is available to be contacted via telephone 7 days per week - For 6 months, monthly telephone follow-up calls are conducted by the APGN. <u>ExN-HaT arm</u>: *Pre-discharge*: - Within 72 hours of admission, an APGN will undertake a health assessment, and prepare a comprehensive transitional care plan. This process includes discussing the planned discharge plan with patient's caregivers, doctor, and ward nurses to individualise the plan - Within 72 hours of admission, a physiotherapist will visit the intervention patients to determine the functional capacity of the patient using measures of ADL and performance tests of balance and gait. This information will be used to plan individualised exercise programs designed to improve strength, stability, coordination, endurance, mobility, and improve self confidence with respect to ADL. The exercise prescription will be developed by the patient, caregivers, doctors, and ward nurses. - Physiotherapist will provide a pedometer together with simple verbal and written instructions regards usage and journaling of activity levels - APGN will visit patients 2nd daily thereafter until discharge to establish and implement the program, monitor progress, and modify transitional care plan if required. *Bridging:* None *Post-discharge:* - Within 48 hours post discharge, an APGN provides one home visit. This home visit is undertaken to 1) identify sufficient caregiver support is available at home, 2) assess the home environment is safe, 3) determine the patient has the required medications and dressings (if				

		required), 4) ensure the patient (and caregiver) fully understand their medication and treatment regimes; 5) reinforce and further explain exercise program and use of pedometer and journal in the home, and 6) provide advice and support to the caregiver - Additional home visits are available by APGN if required - An exercise physiologist conducts 6 weekly visits to reassess the patient's physical measures and functional capacity, evaluate progress with the exercise program, and reset program goals accordingly - Over the initial 4 week period after discharge, a APGN provides weekly telephone follow-up calls - APGN is available to be contacted via telephone 7 days per week - For 6 months, monthly telephone follow-up calls are conducted by the APGN				
Koehler et al.[26] **2009**	Early Care Bundle or Supplemental Care Bundle	Pre-discharge: By Study Care Coordinator - Daily condition specific education - Additional time to identify and address discharge barriers - Extra discharge teaching with focus on self-management By Study Pharmacist: - Medication reconciliation at admit - Daily medication review and education - Additional medication change recommendations as felt to be indicated - Medication reconciliation at discharge - Counselling on discharge Structured documents: - personal health record (completed by the patient and family for future use) Bridging: Structured documents: - Supplemental discharge form given to patient and faxed to PCP Post-discharge: By Study Care Coordinator:	Care coordinators (CC) Clinical pharmacists (CP)	1 of 2 hospital-medicine groups at the 900-bed Baylor University Medical Center, Texas	Max 24 hours after enrolment to 1 week after discharge	Usual care: - Needs assessment - discharge planning - Medication reconciliation at admit - Daily direct patient care - Discharge medication education - Medication reconciliation at discharge - Discharge education - Medication review with new orders and as needed - Medication change recommendations - Standard BUMC discharge form - New discharge prescriptions

		- Follow up call at 5-7 days post-discharge By Study Pharmacist: - Follow up call at 5-7 days post-discharge				
Lin et al.[31] **2015**	Integrated Care and Discharge Support for elderly patients (ICDS)	Pre-discharge: - Risk stratification - Comprehensive geriatric assessment - Discharge planning are performed Bridging: None Post-discharge: Patients receive either: - ICM Case Management or - HST service *ICM Case Management* : - Each high-risk older patient is followed up by a case manager for a period of around 3 months following hospital discharge - Case managers provide home visits and telephone support and they are responsible for community service coordination and ensuring patient compliance with planned services and management. *HST service*: - Rapid and intensive community support for discharged patients, offering services such as meal delivery, household cleaning, respite care, and home assessment and modification Link nurses, ICM case managers, and the HST hold weekly multidisciplinary case conferences chaired by an ICM geriatrician. If needed, referral for rehabilitation in a geriatric day hospital, fast track clinic, or early specialist clinic follow-up can be offered to patients.	At hospital: - Link nurses - Geriatricians Community ICM Case Management: - Social workers - Physiotherapist (PT) - Occupational therapist (OT) - Advanced practice nurse (APN) Community HST service: - Nurses - PT - OT - Other allied health members	Four hospitals of Hong Kong West Cluster.	ICM case management: - From recruitment until 3 months after discharge HST service: - NR	NR
Nielsen et al.[28] **2018**	Elderly Activity Performance Intervention	Pre-discharge: *Component 1:* -Assessment of the patients' performance of daily activities - Assessment of Motor and Process Skills Based on the results in component 1, patients with limitations in performing daily activities	Occupational therapist (OT) Physiotherapist	Short-stay unit (ED) at a 1,150-bed university hospital in Denmark	From enrolment until the day after discharge	Usual practice: relevant medical treatment and care. Referral to occupational therapy and physiotherapy took place only if the medical or nursing

		received components 2 and 3 *Component 2:* - A rehabilitation plan was prescribed. The rehabilitation plan included a description of the patient's previous and current performance of daily activities and specified the need for further rehabilitation. <u>Bridging</u>: - Primary care was informed about the discharge, and visitation of the patient to further rehabilitation <u>Post-discharge</u>: *Component 3:* - Home visit by an OT the day after discharge for patients with a prescribed rehabilitation plan. The home visit aimed to enhance the patient's performance of daily activities and to start rehabilitation. - OT screened the home for safety risks and factors that potently could limit the performance of daily activities.				staff considered it necessary. If the occupational therapist was summoned to asses a patient, the occupational therapist performed a short interview and a non-standardized observation of the patient's performance of basic daily activities and the results were communicated to the municipality homecare staff. If necessary, nurses from the ED organized referral to nursing home care after discharge
Robinson et al.[32] **2015**	Integrated Transition of Care (ITC)	<u>Pre-discharge</u>: - Nutrition screening, and if necessary referral to a dietitian - Allied health review - Discharge medicines reconciliation and patient education by a pharmacist. <u>Bridging</u>: - Records were communicated to primary care providers electronically and via mail, and when required, by telephone. <u>Post-discharge</u>: Provided only to patients who were returning to independent community living - 1 and 3 days after discharge: a telephone assessment, education, and support. The aim was to identify problems not dealt with during the discharge process, assist with self-management, and ensure appropriate health and social supports.	Community nurses A geriatrician A pharmacist Cultural support workers	Two acute general hospitals with 536 and 250 beds in Auckland, New Zealand.	From recruitment until 3 days after discharge	NR

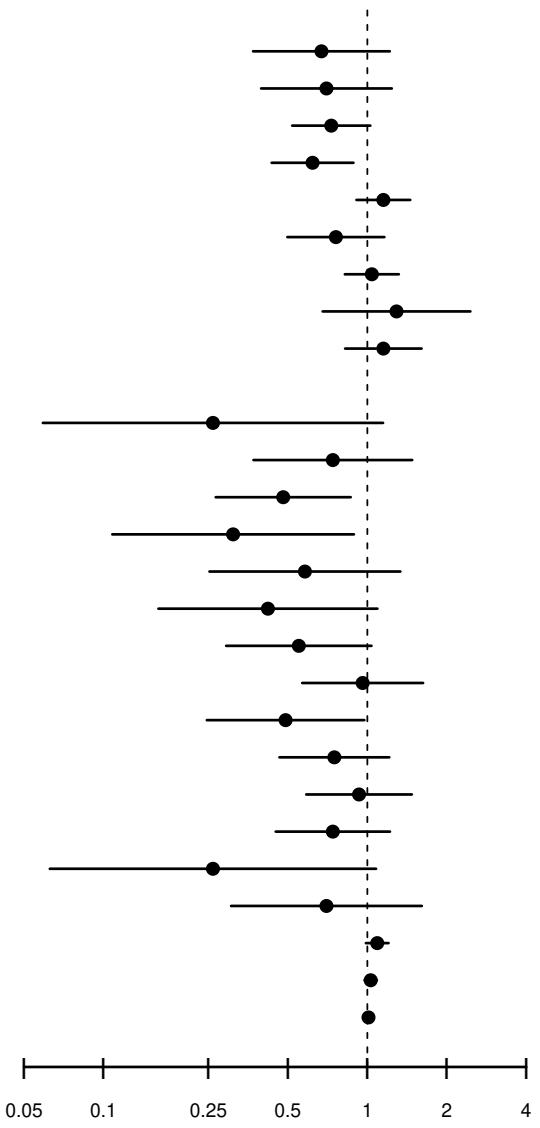
Rottman-Sagebiel et al.[29] **2018**	The Geriatrics Medication Education at Discharge project (GMED)	Pre-discharge: - An individual, face-to-face meeting between the CPS and the patient. - Medication reconciliation - Medication education regarding administration and usage of the patient's medications. - The caregiver, if any, was included in the discussion either at the bedside or by telephone following the face-to-face visit with the patient. - Discuss findings with inpatient team in person or by telephone and through documentation in the CPRS. Bridging: - All information the telephone encounter was communicated to the PCP through CPRS documentation and by telephone as needed. Post-discharge: - A telephone visit within 2 to 3 days of discharge. The purpose of the telephone visit was to perform medication reconciliation, identify and rectify medication errors, provide further patient education, and assist in facilitating appropriate follow-up by the patient's PCP, if required.	Primary care provider (PCP) A clinical pharmacist specialist (CPS) with oversight from a senior CPS with geriatric pharmacology expertise An internist or geriatrician	Veterans Memorial Hospital of the South Texas Veterans Health Care System (STVHCS)	From recruitment until 2 or 3 days after discharge	NR
Sahota et al.[27] **2017**	CIRACT	Pre-discharge: - Comprehensive assessment - Formed a rehabilitation plan - Treated daily (7 days a week, if appropriate) and the duration of rehabilitation they received depended on their needs. - The team liaised with the participant and their carer(s) to visit the participant's home to assess and provide recommendations for equipment and make adaptations and/or modifications as required Bridging: None Post-discharge: - Team visited the participant at home to assess the level of rehabilitation required	Senior occupational therapist (transition coach) Senior physiotherapist Assistant practitioner A social services practitioner	Queen's Medical Centre: a 1,800-bed hospital in Nottingham, UK.	NR	The standard THB-Rehab service was provided on weekdays only. An assessment of the participant's ability to perform certain tasks and provided recommendations for rehabilitation. Referral to community based services for provision of equipment at home, personal care and ongoing rehabilitation in community services where appropriate at

		- Further follow-up visits as deemed necessary - Appropriate referral to additional community services				discharge.
Voss et al.[30] **2011**	Care Transitions Intervention (CTI)	Pre-discharge: - Meet with the patient to establish initial rapport - Introduce the personal health record, including their main health problems, their medications, and questions for their health care providers - Arrange a home visit, ideally within 48 to 72 hours after hospital discharge Bridging: - Discuss topics related to participants with their outpatient providers Post-discharge: *Home visit within 3 days after discharge*: - Reconcile all of the patient's medication regimens - TC imparted skills for effectively communicating care needs during subsequent encounters with health care professionals. - TC also reviewed with the patient any red flags that indicated a condition was worsening and provided education about the initial steps to take to manage the red flags and when to contact the appropriate health care professional. *Phone call*: - Telephoning 3 times during a 28-day post-hospitalization discharge period. The first telephone call within 7 to 10 days generally focused on determining whether the patient had received appropriate services. In the 2 subsequent telephone calls, TC reviewed the patient's progress toward goals established during the home visit, discussed any encounters that took place with other health care professionals, reinforced the importance of maintaining and sharing the personal health record, and supported the patient's role in chronic illness self-management. The final telephone call was conducted by day 30	Transition coach (TC) who had a background in nursing or social work	6 Rhode Island acute care hospitals, including 2 community hospitals, 3 teaching hospitals, and a tertiary care center and teaching hospital, ranging from 129 beds to 719 beds	From enrolment until 30 days after discharge	From randomization to the day of discharge

NR: not reported, CGA: comprehensive geriatric assessment, NH: nursing home, GP: general practitioner, ADL: activity of daily living, PCP: primary care provider, ICM: integrated care model, HST: home support team, ED: emergency department.

Supplementary 6, forest plot grouped by study population

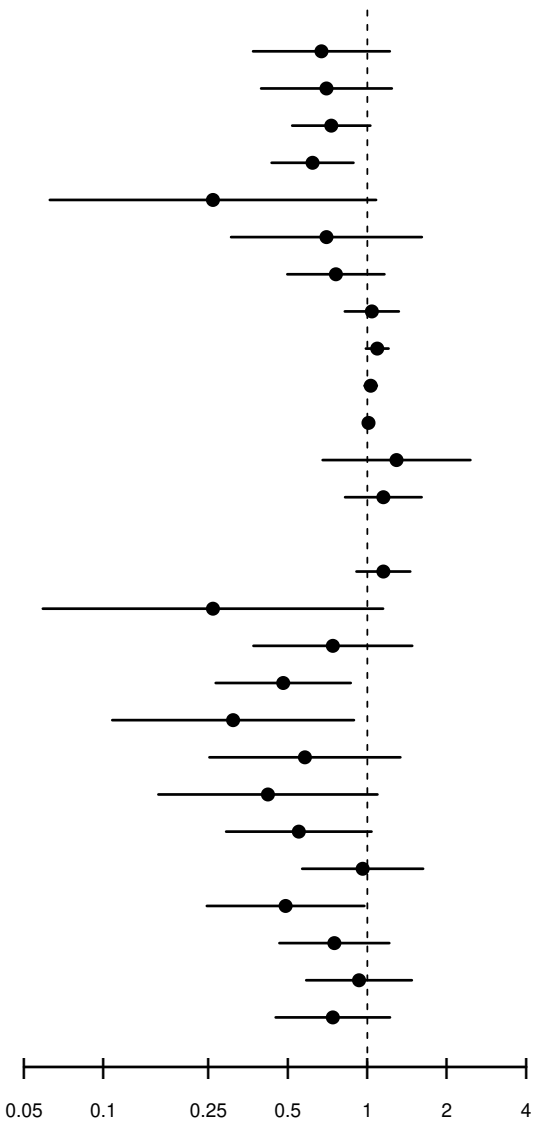
Author	Intervention	OA	RR (95% CI)
Unselected patients			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37–1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40–1.24)
Chow et al.	Home visit	31–91 days	0.73 (0.52–1.03)
Chow et al.	Call	31–91 days	0.62 (0.43–0.89)
Buurman et al.		≥ 92 days	1.15 (0.91–1.45)
Nielsen et al.		≤ 30 days	0.76 (0.50–1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82–1.32)
Sahota et al.		≤ 30 days	1.29 (0.68–2.46)
Sahota et al.		31–91 days	1.15 (0.82–1.61)
Patients at risk			
Courtney et al.		≤ 30 days	0.26 (0.06–1.15)
Courtney et al.		31–91 days	0.74 (0.37–1.48)
Courtney et al.		≥ 92 days	0.48 (0.27–0.87)
Finlayson et al.	EN–Hat	≤ 30 days	0.31 (0.11–0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25–1.33)
Finlayson et al.	N–Hat	≤ 30 days	0.42 (0.16–1.09)
Finlayson et al.	EN–Hat	31–91 days	0.55 (0.29–1.04)
Finlayson et al.	Exercise	31–91 days	0.96 (0.57–1.63)
Finlayson et al.	N–Hat	31–91 days	0.49 (0.25–0.97)
Finlayson et al.	EN–Hat	≥ 92 days	0.75 (0.46–1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59–1.47)
Finlayson et al.	N–Hat	≥ 92 days	0.74 (0.45–1.22)
Koehler et al.		≤ 30 days	0.26 (0.06–1.08)
Koehler et al.		31–91 days	0.70 (0.30–1.61)
Robinson et al.		≤ 30 days	1.09 (0.99–1.20)
Robinson et al.		≤ 30 days	1.03 (0.98–1.09)
Robinson et al.		31–91 days	1.01 (0.97–1.05)



Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

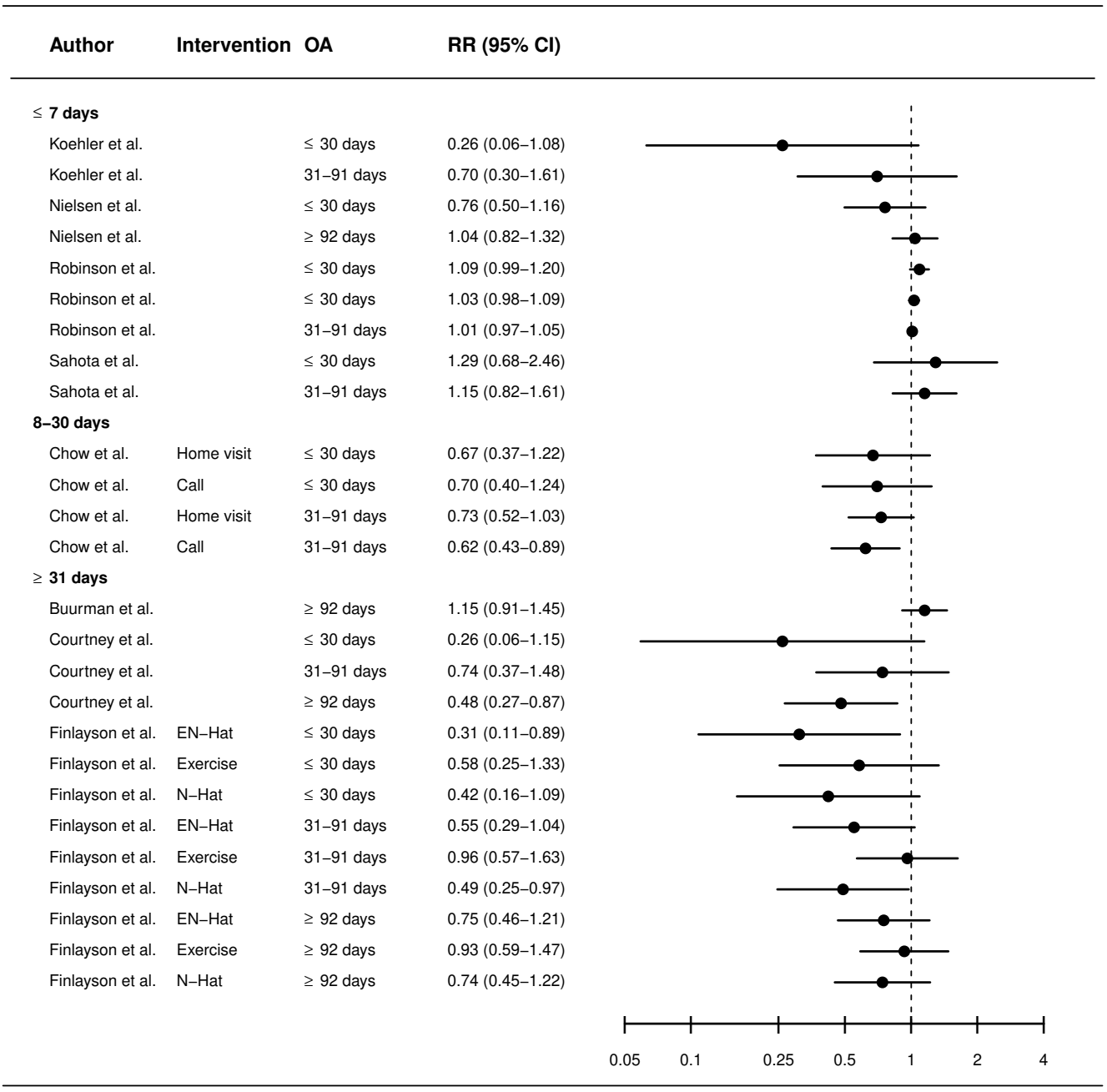
Supplementary 7, forest plot grouped by intervention intensity

Author	Intervention	OA	RR (95% CI)
Low intensity			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37–1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40–1.24)
Chow et al.	Home visit	31–91 days	0.73 (0.52–1.03)
Chow et al.	Call	31–91 days	0.62 (0.43–0.89)
Koehler et al.		≤ 30 days	0.26 (0.06–1.08)
Koehler et al.		31–91 days	0.70 (0.30–1.61)
Nielsen et al.		≤ 30 days	0.76 (0.50–1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82–1.32)
Robinson et al.		≤ 30 days	1.09 (0.99–1.20)
Robinson et al.		≤ 30 days	1.03 (0.98–1.09)
Robinson et al.		31–91 days	1.01 (0.97–1.05)
Sahota et al.		≤ 30 days	1.29 (0.68–2.46)
Sahota et al.		31–91 days	1.15 (0.82–1.61)
High intensity			
Buurman et al.		≥ 92 days	1.15 (0.91–1.45)
Courtney et al.		≤ 30 days	0.26 (0.06–1.15)
Courtney et al.		31–91 days	0.74 (0.37–1.48)
Courtney et al.		≥ 92 days	0.48 (0.27–0.87)
Finlayson et al.	EN–Hat	≤ 30 days	0.31 (0.11–0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25–1.33)
Finlayson et al.	N–Hat	≤ 30 days	0.42 (0.16–1.09)
Finlayson et al.	EN–Hat	31–91 days	0.55 (0.29–1.04)
Finlayson et al.	Exercise	31–91 days	0.96 (0.57–1.63)
Finlayson et al.	N–Hat	31–91 days	0.49 (0.25–0.97)
Finlayson et al.	EN–Hat	≥ 92 days	0.75 (0.46–1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59–1.47)
Finlayson et al.	N–Hat	≥ 92 days	0.74 (0.45–1.22)



Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

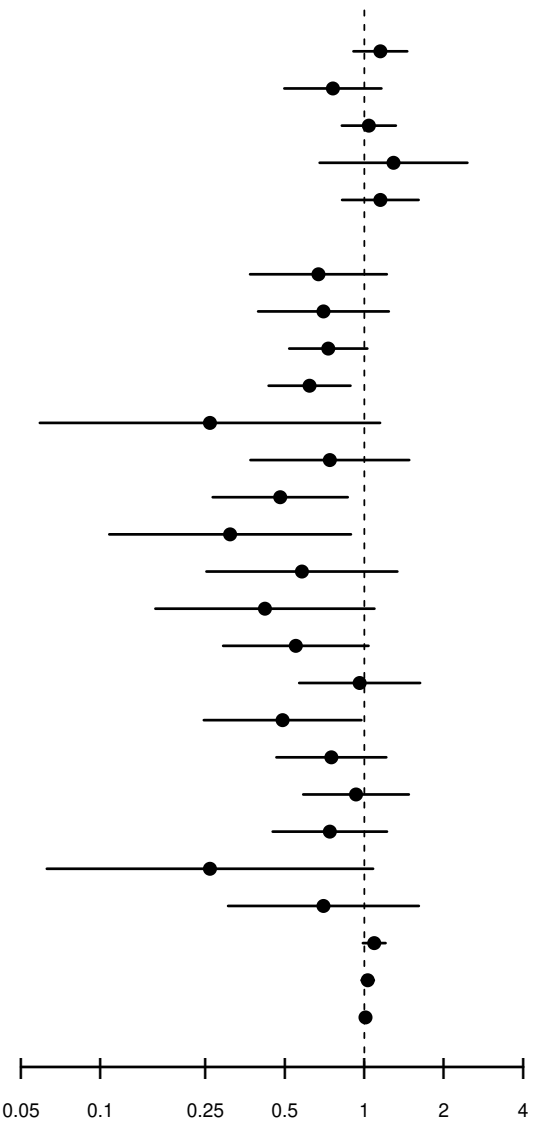
Supplementary 8, forest plot grouped by length of support



Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

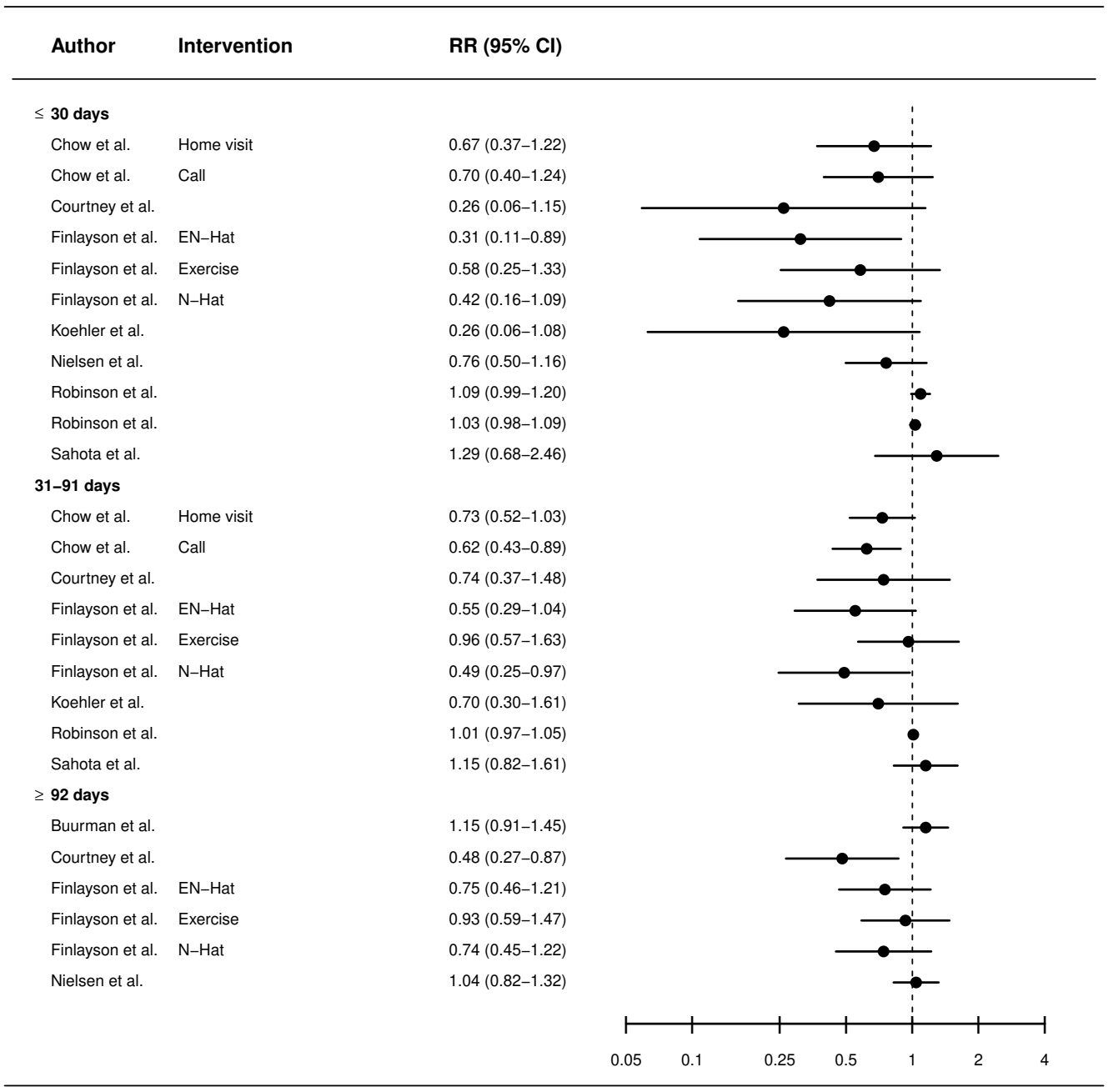
Supplementary 9, forest plot grouped by country of origin

Author	Intervention	OA	RR (95% CI)
European studies			
Buurman et al.		≥ 92 days	1.15 (0.91–1.45)
Nielsen et al.		≤ 30 days	0.76 (0.50–1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82–1.32)
Sahota et al.		≤ 30 days	1.29 (0.68–2.46)
Sahota et al.		31–91 days	1.15 (0.82–1.61)
Non–European studies			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37–1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40–1.24)
Chow et al.	Home visit	31–91 days	0.73 (0.52–1.03)
Chow et al.	Call	31–91 days	0.62 (0.43–0.89)
Courtney et al.		≤ 30 days	0.26 (0.06–1.15)
Courtney et al.		31–91 days	0.74 (0.37–1.48)
Courtney et al.		≥ 92 days	0.48 (0.27–0.87)
Finlayson et al.	EN–Hat	≤ 30 days	0.31 (0.11–0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25–1.33)
Finlayson et al.	N–Hat	≤ 30 days	0.42 (0.16–1.09)
Finlayson et al.	EN–Hat	31–91 days	0.55 (0.29–1.04)
Finlayson et al.	Exercise	31–91 days	0.96 (0.57–1.63)
Finlayson et al.	N–Hat	31–91 days	0.49 (0.25–0.97)
Finlayson et al.	EN–Hat	≥ 92 days	0.75 (0.46–1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59–1.47)
Finlayson et al.	N–Hat	≥ 92 days	0.74 (0.45–1.22)
Koehler et al.		≤ 30 days	0.26 (0.06–1.08)
Koehler et al.		31–91 days	0.70 (0.30–1.61)
Robinson et al.		≤ 30 days	1.09 (0.99–1.20)
Robinson et al.		≤ 30 days	1.03 (0.98–1.09)
Robinson et al.		31–91 days	1.01 (0.97–1.05)



Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

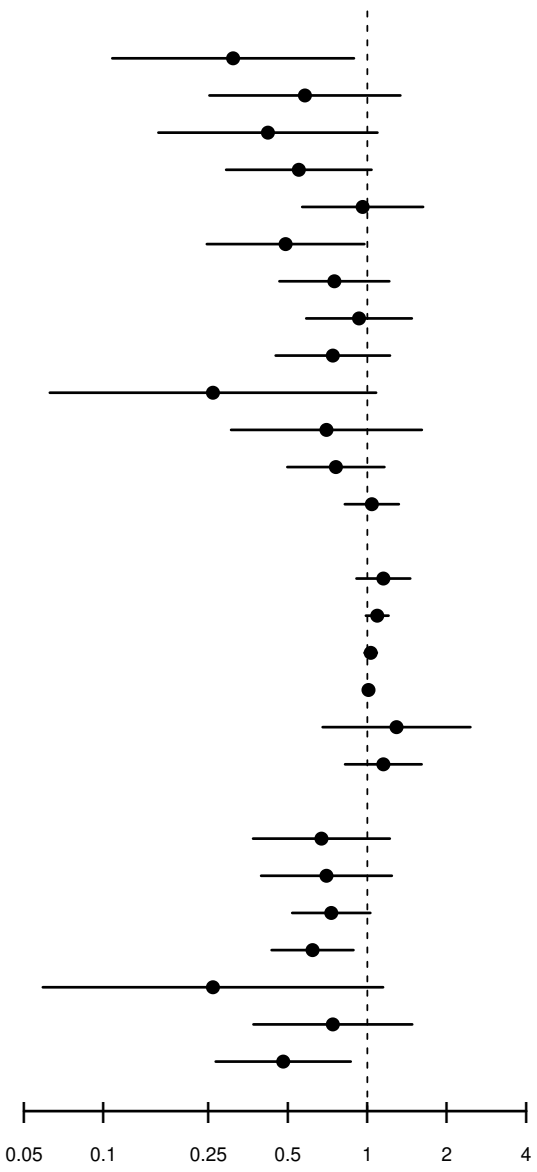
Supplementary 10, forest plot grouped by outcome assessment



Abbreviations: RR, relative risk; CI, confidence interval

Supplementary 11, forest plot grouped by quality assessment

Author	Intervention	OA	RR (95% CI)
Weak			
Finlayson et al.	EN−Hat	≤ 30 days	0.31 (0.11−0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	≤ 30 days	0.42 (0.16−1.09)
Finlayson et al.	EN−Hat	31−91 days	0.55 (0.29−1.04)
Finlayson et al.	Exercise	31−91 days	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	31−91 days	0.49 (0.25−0.97)
Finlayson et al.	EN−Hat	≥ 92 days	0.75 (0.46−1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	≥ 92 days	0.74 (0.45−1.22)
Koehler et al.		≤ 30 days	0.26 (0.06−1.08)
Koehler et al.		31−91 days	0.70 (0.30−1.61)
Nielsen et al.		≤ 30 days	0.76 (0.50−1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82−1.32)
Moderate			
Buurman et al.		≥ 92 days	1.15 (0.91−1.45)
Robinson et al.		≤ 30 days	1.09 (0.99−1.20)
Robinson et al.		≤ 30 days	1.03 (0.98−1.09)
Robinson et al.		31−91 days	1.01 (0.97−1.05)
Sahota et al.		≤ 30 days	1.29 (0.68−2.46)
Sahota et al.		31−91 days	1.15 (0.82−1.61)
Strong			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37−1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40−1.24)
Chow et al.	Home visit	31−91 days	0.73 (0.52−1.03)
Chow et al.	Call	31−91 days	0.62 (0.43−0.89)
Courtney et al.		≤ 30 days	0.26 (0.06−1.15)
Courtney et al.		31−91 days	0.74 (0.37−1.48)
Courtney et al.		≥ 92 days	0.48 (0.27−0.87)



Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval